

Fear of Temperature: Historical Lexical Discovery and Anchor Validation

Executive Summary, Method, and Anchor Overview

This first research round reconstructs an **English-language candidate lexicon** for six historical positions without using frequency, Ngram measurement, co-occurrence, semantic networks, sentiment scoring, or quantitative trend comparison. The central finding is that the six positions do not represent six stages of one stable emotion. Rather, they expose different **lexical conditions of possibility** for talking about temperature, climate, affect, and danger.

The evidence supports a markedly different role for each position. **1842 is defensible as a contextual environmental-history and meteorological lexical baseline, not as an origin point for anthropogenic climate change or climate fear.** Around that date, English-language records strongly support observational terms such as *temperature*, *heat*, *high temperature*, *meteorological observations*, *climate*, and *weather*, together with concrete heat-harm formulations. They do **not** provide comparably strong evidence for a coherent affective vocabulary directed at climatic warming. Contemporary meteorology was still largely observational and descriptive, and historical work on the 1842 European drought likewise places the year within an early-instrumental observation regime rather than a modern climate-change discourse. ¹

The **1938–1949 bridge is historically useful, but 1938 is the strongest specific year.** Guy Stewart Callendar's 1938 paper directly joined *artificial production of carbon dioxide* to an *influence on temperature*, estimated an increase in mean temperature associated with added CO₂, and compared this with observed changes in world temperatures. His 1949 paper subsequently used *climatic change*, *climate*, *temperature rise*, *surface temperatures*, and *human agency*. Modern histories of climate science treat this work as an important reactivation of the CO₂–climate problem and a step toward more physically causal climate science. Crucially, however, Callendar's lexicon is substantially stronger in Layers A and B than in fear, danger, or catastrophe vocabulary. ²

1988 is a strong institutional anchor. The UN General Assembly was already using *climate change*, *global climate patterns*, *greenhouse gases*, *global warming*, *concern*, *threatening*, *severe consequences*, *disastrous*, and *detrimental effects*, while the Toronto Conference framed atmospheric change through *global security*, a *critical problem*, and extreme consequences. WMO and UNEP created the IPCC in 1988, giving this vocabulary an institutional assessment framework. ³

The **2006–2007 bridge should be retained as a period rather than reduced to one year.** In 2006, the Stern Review centrally organized climate change through *risk*, *serious impacts*, *dangerous delay*, and *worst impacts*, while Al Gore's public communication explicitly employed *climate crisis*. In 2007 the IPCC Fourth Assessment cycle consolidated scientific language including the unequivocal warming of the climate system, while the Nobel recognition of the IPCC and Gore explicitly rewarded the construction and dissemination of knowledge concerning human-caused climate change. The combination makes the two-year interval more useful than either date alone for this thesis. ⁴

2015 is a strong temperature-threshold/governance anchor. The Paris Agreement made *global average temperature, well below 2°C, 1.5°C, pre-industrial levels*, and the *threat of climate change* unusually important lexical objects in formal international governance; its associated framework also uses *risk, adverse effects, loss and damage*, and *vulnerability*. ⁵

2022 is a strong contemporary endpoint with a much more explicit affective vocabulary. WMO material from 2022 uses *extreme heatwaves, accumulated heat, greenhouse gas concentrations*, and global-temperature language, while climate-psychology literature available by this period explicitly contains *climate anxiety, eco-anxiety, worry, psychological distress, afraid*, and *hopelessness*. These terms should not be projected backwards. ⁶

The resulting recommended timeline is therefore:

1842 — contextual environmental-history lexical baseline → 1938 — CO₂/temperature scientific bridge → 1988 — institutional climate-risk anchor → 2006–2007 — public-communication/threat-framing bridge → 2015 — temperature-threshold governance anchor → 2022 — contemporary corpus endpoint.

Conceptual and methodological boundary

For this round, **“Fear of Temperature” is not treated as the name of a transhistorical emotion.** The object is a documentary vocabulary through which temperature and heat could be observed; climate could be conceptualised; affective reactions could be expressed; and danger or risk could be attributed. Historical-emotions scholarship cautions against assuming that apparently familiar emotion words carried stable social and affective meanings across time. ⁷

The four layers are therefore analytically separated:

Layer A records physical or observational temperature language.

Layer B records climatic, meteorological, atmospheric, or causal framing.

Layer C records affect vocabulary only where an affective interpretation is at least plausible.

Layer D records danger, harm, threat, risk, or adverse-consequence framing.

A Layer-C term such as *concern* is not automatically equivalent to *fear*. A Layer-D term such as *risk* is not itself evidence that somebody felt afraid. Conversely, the absence of the modern expression *climate anxiety* in an early period is not evidence that people lacked emotional responses to heat or weather. Historical work on the term *climate* itself likewise shows that climate vocabulary has a history and cannot be assumed to be semantically invariant. ⁸

No lexical candidate below was selected because of frequency. Google Books Ngram is mentioned only because the present Viewer interface supports an end year of **2022**; no Ngram query result, count, chart, percentage, or comparative frequency was used in candidate construction. ⁹

Anchor overview

Anchor	Status	Historically defensible function	Evidence strength	Main source types
1842	Primary exploratory	Early meteorological/ environmental lexical baseline; observational temperature and embodied heat-harm language	Strong A/B; mixed D; weak C	Meteorological books/ observations, newspapers, historical climatology
1938–1949	Candidate bridge	CO ₂ -temperature causal framing and renewed carbon-dioxide climate theory	Strong A/B; weak C/D	Callendar papers, history of climate science
1988	Primary	Institutionalisation of climate-change assessment and explicit concern/threat language	Strong A/B/ D; selective C	UN resolution, Toronto declaration, WMO/ IPCC records
2006–2007	Candidate bridge	Public communication plus explicit risk/crisis/danger framing alongside AR4	Strong A/B/ D; mixed C	Stern Review, IPCC AR4, Gore communication, Nobel records
2015	Primary	Formal temperature thresholds and global threat/ risk governance lexicon	Strong A/B/ D; weak-to-mixed C	Paris Agreement/ UNFCCC, contemporary public-health discourse
2022	Primary endpoint	Extreme-heat/climate-risk vocabulary plus explicitly named climate-related affect	Strong across A–D	WMO, IPCC, climate-psychology scholarship

Anchor — 1842

Historical context. 1842 works best when separated from later anthropogenic-warming narratives. Luke Howard's *Climate of London* had already established an English meteorological vocabulary involving *climate*, *temperature*, *thermometer*, atmospheric conditions, seasons, and systematic observation. A contemporary December 1842 meteorological item described a notably high temperature relative to a mean, showing that temperature anomalies could be articulated observationally without implying a modern climate-change framework. ¹⁰

The year itself also offers environmental stress contexts. Modern historical climatology identifies the 1842 European drought as occurring before national meteorological-service networks were fully developed and finds substantial documentary reporting of drought and heat impacts, including illness. Importantly for an English lexicon, an 1842 colonial newspaper record described sheep as “too hot and close” during shipment and stated that heat from close stowage caused deaths. This supplies concrete heat-harm wording without requiring any inference of “climate anxiety.” ¹¹

Evidence key. **E1842-1** = Howard, *Climate of London*, 2nd ed. (1833). ¹² **E1842-2** = “Meteorological observations for December 1842,” contemporary meteorological publication; author/volume/page metadata were not sufficiently exposed in the accessible record and are therefore not invented here.

¹³ **E1842-3** = *New Zealand Gazette and Wellington Spectator*, “Colonial Extracts,” 22 October 1842. ¹⁴ **E1842-4** = Brázdil et al. on the 1842 European drought. ¹⁵ **E1842-5** = Barry’s historical study of *climate/climatology*. ¹⁶ **E1842-6** = historical study of nineteenth-century medical meteorology; used only contextually because some underlying evidence is non-English. ¹⁷ For failed affect probes, “negative validation” means only that no sufficiently strong temperature-linked English attestation was established in the reviewed evidence set; it does **not** assert universal absence from all nineteenth-century print.

Thirty candidate records — exactly 8 A, 8 B, 7 C, 7 D.

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1842-A-01	1842	temperature	A	unigram	—	1833–1842	E1842-1,2	Core meteorological variable
1842-A-02	1842	mean temperature	A	phrase	average temperature	1833–1842	E1842-1,2	Used in observational comparison
1842-A-03	1842	high temperature	A	phrase	elevated temperature	1842	E1842-2	Contemporary observational description of unusual reading
1842-A-04	1842	heat	A	unigram	—	1842	E1842-3,4	Environmental and bodily heat explicitly described
1842-A-05	1842	great heat	A	phrase	severe heat	around 1842	E1842-4	Modern historical synthesis phrase documenting condition
1842-A-06	1842	oppressive heat	A	phrase	severe felt heat	1840s probe	E1842-1–4 validation set	Period-plausible expression not successfully validated in high-quality English evidence

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1842-A-07	1842	unusual heat	A	phrase	anomalous heat	1840s probe	E1842-1-4 validation set	Historical relevant plausible exact high quality attestation established
1842-A-08	1842	too hot and close	A	historical phrase	excessive enclosed heat	1842	E1842-3	Contemporary shipping evidence describing animals becoming hot in cargo stowage
1842-B-01	1842	climate	B	unigram	—	1833	E1842-1,5	Established meteorological geographical term
1842-B-02	1842	meteorology	B	unigram	—	1830s–1840s	E1842-1	Established scientific label
1842-B-03	1842	meteorological observations	B	phrase	systematic weather observation	1833–1842	E1842-1,2	Explicitly practiced period
1842-B-04	1842	weather	B	unigram	—	1842	E1842-3	Ordinary environmental description
1842-B-05	1842	atmosphere	B	unigram	atmospheric conditions	1833–1842	E1842-1	Standard meteorological framing
1842-B-06	1842	seasons	B	unigram	seasonal climate	1833	E1842-1	Part of Howard's climatology observation scheme

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1842-B-07	1842	climatic influences	B	phrase	environmental climatic influence	nineteenth-century context	E1842-6	Historical medical meteorology scholarship documents expressing underlying context include English evidence
1842-B-08	1842	change of climate	B	historical phrase	climatic change	nineteenth-century usage	E1842-5	Terminology history support "change of climate" construction
1842-C-01	1842	fear	C	unigram	fear	1842	E1842-3	Contemporary newspaper weather adjacent but object damage, rudder, temper
1842-C-02	1842	alarm	C	unigram	alarm/fear	not temperature-validated	E1842-1-3 negative validation	No security heat/climate affective establishment
1842-C-03	1842	anxiety	C	unigram	anxiety	not temperature-validated	E1842-1-3 negative validation	No security temper linked u establishment
1842-C-04	1842	apprehension	C	unigram	fearful anticipation	not temperature-validated	E1842-1-3 negative validation	No security heat/climate affect attestation establishment
1842-C-05	1842	dread	C	unigram	dread	not temperature-validated	E1842-1-3 negative validation	No strong temper specific evidence reviewed

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1842-C-06	1842	distress	C	unigram	distress	1842 false-positive	E1842-3	Newspaper uses <i>distress</i> in economic social context; illustration of false-positive danger
1842-C-07	1842	relief	C	unigram	affective/physical relief	1840s exploratory	E1842-3,4	Relief from adverse conditions; concept plausible; evidence strong; to equal fear
1842-D-01	1842	danger	D	unigram	danger	not heat-validated	E1842-1-4 negative validation	Generic word; not strong; elevated temperature use established here
1842-D-02	1842	serious importance	D	historical phrase	serious consequence	1842	E1842-3	Delay until hot, conditions transported; condition described; serious importance
1842-D-03	1842	illness	D	unigram	health harm	1842 context	E1842-4	Historical climatology links 1842 drought reported
1842-D-04	1842	death	D	unigram	mortality	1842	E1842-3	Heat in stowage explicitly to animal deaths

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1842-D-05	1842	loss	D	unigram	material/life loss	1842	E1842-3	Shipping repeated framed consequences as loss
1842-D-06	1842	damage	D	unigram	harm/damage	1842	E1842-3	Weather related damage appears same newspaper environment
1842-D-07	1842	heat ... caused the death	D	historical phrase	heat-caused mortality	1842	E1842-3	Direct construction attributed sheep mortality to heat stowage

Weak/problematic candidates. The most important negative result is the weakness of Layer C. *Fear*, *anxiety*, *alarm*, *dread*, and *apprehension* were certainly available English emotion words, but this round did not establish sufficiently strong evidence that they formed a recognisable temperature/climate affect lexicon around 1842. That distinction matters: historical availability of an emotion word is not the same as historical availability of a concept equivalent to modern climate fear. The same caution applies to *great heat* and *unusual heat* when a modern historian's English description might be mistaken for exact English wording in an original, sometimes non-English, document. ¹⁸

What does 1842 actually represent? It represents, most defensibly, an **early lexical intersection of meteorological observation, climatic description, unusual/elevated temperature, and concrete environmental heat harm**. It can also accommodate affective investigation, but the reviewed record does not justify making affect the defining historical feature of the year. The dramatic 1842 European drought is useful context but is not itself a reason to reinterpret the date as an anthropogenic-climate milestone. ¹⁵

Verdict: CONTEXTUAL ENVIRONMENTAL-HISTORY BASELINE. Retain 1842, but explicitly describe it as an exploratory baseline before modern anthropogenic climate-change vocabulary, not as an origin of climate change, global warming, or fear of temperature.

Candidate Bridge — 1938–1949

Historical context. Callendar's 1938 paper is unusually valuable lexically because its title itself brings three elements into one scientific proposition: human-produced CO₂ (*artificial production*), an atmospheric agent (*carbon dioxide*), and an outcome (*influence on temperature*). Its abstract estimates an increase in mean temperature attributable to artificially produced carbon dioxide and compares that result with temperature observations from meteorological stations showing increased world temperatures. ¹⁹

The 1949 paper *Can Carbon Dioxide Influence Climate?* makes the climatic framing still clearer. Callendar describes carbon dioxide as partially opaque to terrestrial heat radiation, discusses a heat-trapping effect, calculates a temperature rise under increased CO₂, considers surface-temperature changes, aridity and glacier recession, and finally turns to whether the changes could be produced by **human agency**. ²⁰

This is therefore a scientific-causal transition, not yet a clear fear transition. Historical scholarship treats Callendar as important in reviving the CO₂ theory when many contemporaries regarded anthropogenic modification of global climate as implausible or poorly established. ²¹

Evidence key. **E38-1** = Callendar 1938. ¹⁹ **E38-2** = Callendar 1949. ²⁰ **E38-3** = Fleming, *The Callendar Effect*. ²² **E38-4** = Weart/AIP historical reconstruction. ²³ “No” candidates below record failed validation of plausible threat/affect probes; they are not claims that the word did not exist anywhere in 1938–1949.

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1938-A-01	1938–49	temperature	A	unigram	—	1938, 1949	E38-1,2	Central dependent physical variable
1938-A-02	1938–49	mean temperature	A	phrase	average temperature	1938	E38-1	Callendar estimated mean-temperature effect
1938-A-03	1938–49	world temperatures	A	phrase	global temperatures	1938	E38-1	Used for observed large scale temperature change
1938-A-04	1938–49	temperature increase	A	phrase	warming	1938	E38-1	Increased temperature associated CO ₂
1938-A-05	1938–49	rise in temperature	A	phrase	warming	1949	E38-2	CO ₂ change connected calculated temperature rise
1938-A-06	1938–49	surface temperatures	A	phrase	surface temperature	1949	E38-2	Surface temperature discussed a response to atmospheric change

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1938-A-07	1938–49	temperature observations	A	phrase	instrumental temperature records	1938	E38-1	Observation station records were compiled
1938-A-08	1938–49	temperature change	A	phrase	warming/cooling	1949	E38-2	CO ₂ variations discussed in relation to change in temperature
1938-B-01	1938–49	carbon dioxide	B	phrase	CO ₂	1938–49	E38-1,2	Central atmospheric agent
1938-B-02	1938–49	atmospheric carbon dioxide	B	phrase	atmospheric CO ₂	1938–49	E38-1–3	CO ₂ abundance concentrated framed atmospheric
1938-B-03	1938–49	artificial production of carbon dioxide	B	historical phrase	anthropogenic CO ₂ emissions	1938	E38-1	Exact conceptual centre of 1 st paper title
1938-B-04	1938–49	atmosphere	B	unigram	—	1938–49	E38-1,2	Medium through which CO ₂ radiative effect is fra
1938-B-05	1938–49	climate	B	unigram	—	1949	E38-2	Appears directly in 1 st paper title
1938-B-06	1938–49	climatic change	B	phrase	climate change	1949	E38-2	Opening discussion frames CO ₂ theory as interpretation of climatic change
1938-B-07	1938–49	carbon dioxide theory	B	historical phrase	CO ₂ climate theory	period context	E38-2–4	Historical scientific discussion treated a C theory of climate

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1938-B-08	1938–49	human agency	B	phrase	anthropogenic influence	1949	E38-2	Callendar explicitly tu to whether changes co arise throu human age
1938-C-01	1938–49	doubtful	C	historical variant	doubt	period scientific debate	E38-2–4	Doubt characteris contempor reception/ epistemic position
1938-C-02	1938–49	uncertainty	C	unigram	uncertainty	period context	E38-3,4	Historical accounts st unresolved scientific uncertainty
1938-C-03	1938–49	concern	C	unigram	concern	not securely validated	E38-1–4 negative validation	No strong period clim affective attestation established
1938-C-04	1938–49	fear	C	unigram	fear	not validated	E38-1–4 negative validation	Callendar materials reviewed d not establish fear lexicon
1938-C-05	1938–49	alarm	C	unigram	alarm	not validated	E38-1–4 negative validation	Not establish as climate-affective wording
1938-C-06	1938–49	anxiety	C	unigram	anxiety	not validated	E38-1–4 negative validation	No strong climate-link use establish
1938-C-07	1938–49	apprehension	C	unigram	apprehension	not validated	E38-1–4 negative validation	No secure climate-link use found

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1938-D-01	1938-49	aridity	D	unigram	drying impact	1949	E38-2	Increased aridity proposed as possible indication of CO ₂ -related change
1938-D-02	1938-49	recession of the glaciers	D	phrase	glacier retreat	1949	E38-2	Proposed among possible indications of climatic influence
1938-D-03	1938-49	danger	D	unigram	danger	not validated	E38-1-4 negative validation	Not established as a defining Callendar-period term
1938-D-04	1938-49	risk	D	unigram	risk	not validated	E38-1-4 negative validation	No strong climate risk formulation established
1938-D-05	1938-49	harmful	D	unigram	harm	not validated	E38-1-4 negative validation	Detrimental framing not securely established
1938-D-06	1938-49	adverse effect	D	phrase	adverse climate impact	not validated	E38-1-4 negative validation	Modern-sounding impact phrase not securely supported
1938-D-07	1938-49	problem	D	unigram	problem/threat	not securely validated	E38-1-4 negative validation	CO ₂ question was scientifically contested, modern “climate problem” framing is not established

Anchor-year evaluation. 1938 is more defensible than 1949 as the bridge date because the 1938 title itself performs the key lexical conjunction: artificial CO₂ production → temperature influence. The 1949 paper is extremely useful corroboration because it explicitly shifts into *climate*, *climatic change*, and *human agency*, but it reads as development of an already articulated programme rather than the best starting marker inside the proposed interval. ²⁴

The period does introduce a configuration not available at the 1842 baseline: **a named atmospheric constituent, a human production mechanism, radiative reasoning, and observed large-scale temperature change are brought into a single causal research problem.** That does not mean 1938 was the “origin of global warming”; nineteenth-century CO₂ theory preceded Callendar, and historical scholarship explicitly situates him within a longer line of work. ²⁵

Verdict: ACCEPT — specific year: 1938. Retain 1949 as an important contextual source inside the bridge window, not as a seventh anchor.

Anchor — 1988

Historical context. In December 1988, UN General Assembly Resolution 43/53 stated that human activities could alter *global climate patterns*, referred to increasing atmospheric concentrations of *greenhouse gases*, *global warming*, potentially *severe economic and social consequences*, effects that could be *disastrous*, *detrimental effects*, and the need to treat climate change as a priority issue. The resolution also endorsed the newly established IPCC. ²⁶

The Toronto Conference on the Changing Atmosphere supplied an even stronger threat lexicon: *climate warming*, changed frequencies of *climatic extremes*, a *critical problem*, increasing seriousness, *global security*, and extreme-consequence framing. WMO confirms that the IPCC was created jointly by WMO and UNEP in 1988. ²⁷

Evidence key. **E88-1** = UNGA Resolution 43/53. ²⁶ **E88-2** = Toronto Conference Declaration. ²⁸ **E88-3** = WMO institutional history of IPCC. ²⁹ **E88-4** = archival/historical reproduction of James Hansen's 1988 testimony, used for temperature and heat-wave terms rather than for claims about public reception. ³⁰

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1988-A-01	1988	global warming	A	phrase	planetary warming	1988	E88-1,4	Explains official scientific
1988-A-02	1988	warming	A	unigram	temperature increase	1988	E88-1,2,4	Used in climate temperature context
1988-A-03	1988	climate warming	A	phrase	climatic warming	1988	E88-2	Toronto declaration uses as climate concept
1988-A-04	1988	global temperature	A	phrase	global mean temperature	1988	E88-4	Hansen's testimony frames large-scale temperature change

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1988-A-05	1988	temperature rise	A	phrase	warming	around 1988	E88-1,4	Warm frame rising temperature
1988-A-06	1988	temperature increase	A	phrase	warming	1988	E88-4	Scientific temperature change forms
1988-A-07	1988	warming trend	A	phrase	observed warming tendency	around 1988	E88-4	Trend language below observed climate discussion
1988-A-08	1988	summer heat waves	A	phrase	extreme heat	1988	E88-4	Heat involvement among expected observed warm related extreme
1988-B-01	1988	climate change	B	phrase	—	1988	E88-1,3	Central IPCC institution term
1988-B-02	1988	climatic change	B	historical variant	climate change	1988	E88-1	Rescue retail change word
1988-B-03	1988	global climate	B	phrase	planetary climate	1988	E88-1	Official frame
1988-B-04	1988	global climate patterns	B	phrase	large-scale climate patterns	1988	E88-1	Human activity capacity alteration pattern
1988-B-05	1988	greenhouse gases	B	phrase	heat-trapping gases	1988	E88-1	Atmospheric growth to global warming

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1988-B-06	1988	atmospheric concentrations	B	phrase	atmospheric GHG concentration	1988	E88-1	Growth in atmospheric concentrations of greenhouse gases (GHGs) has led to a significant increase in the concentration of atmospheric GHGs, which is a major cause of global warming and climate change.
1988-B-07	1988	human activities	B	phrase	anthropogenic influence	1988	E88-1	Human activities, such as the burning of fossil fuels, have led to a significant increase in the concentration of atmospheric GHGs, which is a major cause of global warming and climate change.
1988-B-08	1988	changing atmosphere	B	phrase	atmospheric change	1988	E88-2	Changes in the atmosphere, such as the increase in the concentration of atmospheric GHGs, have led to a significant increase in the concentration of atmospheric GHGs, which is a major cause of global warming and climate change.
1988-C-01	1988	concern	C	unigram	concern	1988	E88-1	Responsible for the increase in the concentration of atmospheric GHGs, which is a major cause of global warming and climate change.
1988-C-02	1988	concerned	C	morphological variant	concern	1988	E88-1	Preparation of the report on the state of the environment, which is a major cause of global warming and climate change.
1988-C-03	1988	alarm	C	unigram	alarm	not securely validated	E88-1,2 negative validation	Strong evidence of the impact of human activities on the environment, but the impact is not yet fully understood.
1988-C-04	1988	fear	C	unigram	fear	not securely validated	E88-1,2 negative validation	Threats to the environment, such as the increase in the concentration of atmospheric GHGs, have led to a significant increase in the concentration of atmospheric GHGs, which is a major cause of global warming and climate change.
1988-C-05	1988	anxiety	C	unigram	anxiety	not validated	E88-1,2 negative validation	No clear evidence of the impact of human activities on the environment, but the impact is not yet fully understood.

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1988-C-06	1988	worry	C	unigram	worry	not validated	E88-1,2 negative validation	Not established selection in institutional texts
1988-C-07	1988	urgency	C	unigram	urgent concern	1988 context	E88-1,2	Time immemorial action is striven though <i>urgent</i> equi fear
1988-D-01	1988	threatening	D	morphological variant	threat	1988	E88-1	Human causation alter desc threat gene
1988-D-02	1988	severe consequences	D	phrase	serious adverse consequences	1988	E88-1	Economic social consequences desc pote severe
1988-D-03	1988	disastrous	D	unigram	disaster-level harm	1988	E88-1	Possible effects expli char as di
1988-D-04	1988	detrimental effects	D	phrase	adverse effects	1988	E88-1	Rescue calls prev detri clima
1988-D-05	1988	critical problem	D	phrase	severe climate problem	1988	E88-2	Toronto declar char char atm prob critic

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
1988-D-06	1988	global security	D	phrase	planetary/ security threat	1988	E88-2	Toronto conference explicitly concerned atmospheric and
1988-D-07	1988	ultimate consequences	D	phrase	long-term severe consequences	1988	E88-2	Toronto state used high consequences framing

Major ambiguity issues. The strongest Layer-C term is *concern*, but UN diplomatic language makes its emotional status ambiguous: “concerned” can function as an institutional stance rather than evidence of individual fear. *Urgency* is even less securely affective. By contrast, Layer D is unmistakably developed: *threatening*, *severe consequences*, *disastrous*, *critical problem*, and *global security* can be validated without converting them into claims about fear. ³¹

This makes 1988 especially valuable for Round 2: it offers a documentary setting in which **risk/threat framing and affect-adjacent language coexist but remain analytically separable**.

Candidate Bridge — 2006–2007

Historical context. The Stern Review was published in October 2006 and explicitly described itself as assessing climate-change *costs and risks*. Its headline findings said that strong early action could avoid the *worst impacts*, that climate change could have *very serious impacts*, and that delay would be *dangerous* and costlier. These are unusually clear Layer-D lexical anchors. ³²

At almost the same moment, Al Gore's 2006 public communication used *climate crisis* as an organising expression and *global warming* as a public-facing term. ³³ In 2007, the IPCC Fourth Assessment Report's synthesis used the now-famous scientific formulation that warming of the climate system was unequivocal and consolidated vocabulary around observed warming, greenhouse gases, anthropogenic influence and radiative forcing. ³⁴ The 2007 Nobel Peace Prize to the IPCC and Gore explicitly recognised efforts to build and disseminate knowledge about human-caused climate change, while Gore's Nobel-era language used *climate crisis* and strong threat framing. ³⁵

Evidence key. **E06-1** = Stern Review/LSE archival presentation. ³² **E06-2** = IPCC AR4 synthesis materials. ³⁴ **E06-3** = Gore, TED2006, *Averting the Climate Crisis*. ³³ **E07-1** = Nobel/UN records on 2007 award and Gore/IPCC. ³⁵ **E07-2** = Dietz, Hope and Patmore's 2007 scholarly phrase *dangerous climate change*, bibliographically recorded on the Stern Review research page. ³² **E07-3** = contemporary reporting of Gore's *planetary emergency* formulation. ³⁶

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	H
0607-A-01	2006–07	global warming	A	phrase	planetary warming	2006–07	E06-2,3	S
0607-A-02	2006–07	warming	A	unigram	temperature increase	2007	E06-2	A
0607-A-03	2006–07	global average temperature	A	phrase	global mean temperature	2006–07	E06-1,2	C
0607-A-04	2006–07	temperature increase	A	phrase	warming	2006–07	E06-1,2	P
0607-A-05	2006–07	temperature rise	A	phrase	warming	2006–07	E06-1,2	i
0607-A-06	2006–07	warming of the climate system	A	phrase	systemic climatic warming	2007	E06-2	c
0607-A-07	2006–07	heat waves	A	phrase	extreme heat	2006–07	E06-1,2	a
0607-A-08	2006–07	warmer	A	morphological variant	warming	2006–07	E06-1,2	T
0607-B-01	2006–07	climate change	B	phrase	—	2006–07	E06-1,2	c
0607-B-02	2006–07	climate system	B	phrase	Earth climate system	2007	E06-2	t

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	H
0607-B-03	2006–07	greenhouse gases	B	phrase	GHGs	2006–07	E06-1,2	C e v
0607-B-04	2006–07	greenhouse gas concentrations	B	phrase	atmospheric GHG concentration	2007	E06-2	A c c c f
0607-B-05	2006–07	anthropogenic warming	B	phrase	human-caused warming	2007	E06-2	A v
0607-B-06	2006–07	human influence	B	phrase	anthropogenic influence	2007	E06-2	A f t h i
0607-B-07	2006–07	radiative forcing	B	phrase	climate forcing	2007	E06-2	S c v
0607-B-08	2006–07	global climate	B	phrase	planetary climate	2006–07	E06-1,2	L c f
0607-C-01	2006–07	concern	C	unigram	concern	2006–07	E06-1,2	C l c v a
0607-C-02	2006–07	fear	C	unigram	fear	not securely validated in core set	E06-1–3 negative validation	S n r v
0607-C-03	2006–07	worry	C	unigram	worry	not securely validated	E06-1–3 negative validation	M e c t

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	H e
0607-C-04	2006–07	anxiety	C	unigram	anxiety	not securely validated	E06-1–3 negative validation	M c s a
0607-C-05	2006–07	alarm	C	unigram	alarm	not securely validated	E06-1–3 negative validation	C is a a
0607-C-06	2006–07	urgency	C	unigram	urgent concern	2006–07 context	E06-1,3,E07-1	E in a is
0607-C-07	2006–07	awareness	C	unigram	public awareness	2006–07	E07-1	k c P a in in
0607-D-01	2006–07	risk	D	unigram	risk	2006	E06-1	S c a a
0607-D-02	2006–07	dangerous	D	unigram	danger	2006–07	E06-1,E07-2	D c c f a
0607-D-03	2006–07	serious impacts	D	phrase	severe impacts	2006	E06-1	S v in
0607-D-04	2006–07	worst impacts	D	phrase	severe climate impacts	2006	E06-1	A in h f
0607-D-05	2006–07	climate crisis	D	phrase	climate framed as crisis	2006–07	E06-3,E07-1	E c f

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	
0607-D-06	2006–07	planetary emergency	D	phrase	global climate emergency	2007	E07-3	
0607-D-07	2006–07	threat	D	unigram	threat	2007	E07-1	

Anchor-year evaluation. A single-year choice would lose something important. **2006** contains two unusually useful public/policy documents: Stern's systematic *risk/danger/impact* vocabulary and Gore's *climate crisis* communication. **2007** adds AR4's scientific consolidation and the Nobel moment, which explicitly connects scientific knowledge production to public dissemination. ³⁷

This does **not** establish that *climate crisis*, *risk*, or *dangerous climate change* were coined in 2006–2007, nor that their frequency suddenly increased. The narrower conclusion is that the two-year interval supplies a historically coherent documentary cluster in which institutional climate science, economic risk assessment, and high-intensity public threat framing are simultaneously visible.

Verdict: ACCEPT — period: 2006–2007.

Anchor — 2015

Historical context. The Paris Agreement was adopted on 12 December 2015. Its temperature goal is framed as holding the increase in *global average temperature* well below 2°C above *pre-industrial levels* while pursuing efforts to limit the *temperature increase* to 1.5°C. UNFCCC's official account describes this as strengthening the global response to the *threat of climate change*. ⁵

The same official framework contains *anthropogenic emissions*, *greenhouse gas emissions*, *adverse effects*, *loss and damage*, *vulnerability*, and *risk*. Contemporary 2015 public-health communication also explicitly called climate change a *present threat* and described extreme heat as posing danger to health. ³⁸

The affective layer is much less straightforward. *Concern* is present in climate diplomacy, but formulations such as “common concern” have a legal and diplomatic function and cannot automatically be interpreted as an individual's affective state. Likewise, *public awareness* is explicitly present in the Paris architecture but is cognitive/communicative rather than affective. ³⁹

Evidence key. **E15-1** = UNFCCC, *The Paris Agreement*. ⁴⁰ **E15-2** = UNFCCC, *Key aspects of the Paris Agreement*. ³⁹ **E15-3** = contemporary 2015 reporting of White House climate/public-health framing. ⁴¹ **E15-4** = 2015 Lancet Commission on health and climate change. ⁴²

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
2015-A-01	2015	global average temperature	A	phrase	global mean temperature	2015	E15-1,2	Central Paris temperature goal formulation
2015-A-02	2015	temperature increase	A	phrase	warming	2015	E15-1,2	Agreement specifies limiting temperature increase
2015-A-03	2015	global temperature rise	A	phrase	global warming	2015	E15-2	UNFCCC describes global temperature rise
2015-A-04	2015	2°C	A	phrase/ threshold	two-degree warming threshold	2015	E15-1,2	Formal upper temperature goal reference
2015-A-05	2015	1.5°C	A	phrase/ threshold	1.5-degree warming threshold	2015	E15-1,2	Formal aspirational temperature limit
2015-A-06	2015	global warming	A	phrase	planetary warming	2015	E15-1	Official UNFCCC contextual vocabulary
2015-A-07	2015	extreme heat	A	phrase	severe heat	2015	E15-3,4	Public-health discourse treats extreme heat as climate-related consequence
2015-A-08	2015	heat waves	A	phrase	extreme heat events	2015	E15-3,4	Health-imp discourse identifies heat-wave harms
2015-B-01	2015	climate change	B	phrase	—	2015	E15-1,2	Defining subject of Agreement

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
2015-B-02	2015	greenhouse gas emissions	B	phrase	GHG emissions	2015	E15-1,2	Central mitigation vocabulary
2015-B-03	2015	pre-industrial levels	B	phrase	historical temperature baseline	2015	E15-1,2	Defines reference against which limits are measured
2015-B-04	2015	climate system	B	phrase	Earth climate system	2015 context	E15-1,4	Scientific-governance framing
2015-B-05	2015	anthropogenic emissions	B	phrase	human-caused emissions	2015	E15-2	Article 4 framing distinguish anthropogenic sources/removals
2015-B-06	2015	global response	B	phrase	international climate response	2015	E15-2	Agreement aims to strengthen global response to threat
2015-B-07	2015	low GHG emissions	B	phrase	low-emission pathway	2015	E15-2	Official pathway language
2015-B-08	2015	changing climate	B	phrase	climate change	2015	E15-1	Official material used in changing-climate formulation
2015-C-01	2015	concern	C	unigram	concern	2015	E15-1,2	Climate diplomacy contains concern framing
2015-C-02	2015	common concern of humankind	C	historical/legal phrase	shared concern	2015	E15-1	Paris-era legal framing treats climate as common concern

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
2015-C-03	2015	urgency	C	unigram	urgent concern	2015 context	E15-1,2	Accelerated urgent action vocabulary relevant
2015-C-04	2015	public awareness	C	phrase	awareness	2015	E15-2	Article 12 framework explicitly includes public awareness
2015-C-05	2015	fear	C	unigram	fear	not core-validated	E15-1–4 negative validation	Threat language does not validate fear as institutional keyword
2015-C-06	2015	anxiety	C	unigram	anxiety	not core-validated	E15-1–4 negative validation	Climate-psychological anxiety not established central in selected 20 anchor sources
2015-C-07	2015	worry	C	unigram	worry	not core-validated	E15-1–4 negative validation	Not validated as central Paris-era anchor term
2015-D-01	2015	threat of climate change	D	phrase	climate threat	2015	E15-2	UNFCCC explicitly frames global response to the threat
2015-D-02	2015	risk	D	unigram	climate risk	2015	E15-1,2,4	Risk reduction and climate consequences central
2015-D-03	2015	adverse effects	D	phrase	harmful climate effects	2015	E15-1,2	Official Paris framework uses adverse effects language

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
2015-D-04	2015	loss and damage	D	phrase	climate losses/ damages	2015	E15-2	Article 8 framework explicitly names loss and damage
2015-D-05	2015	vulnerability	D	unigram	susceptibility to harm	2015	E15-1,2	Vulnerability reduction is explicit
2015-D-06	2015	danger	D	unigram	danger	2015	E15-3	Extreme heat climate impacts framed as danger to public health
2015-D-07	2015	present threat	D	phrase	current climate threat	2015	E15-3	White House adviser explicitly contrasted present with merely future threat

Major ambiguity issues. The major methodological trap at 2015 is to read formal governance language as emotion. *Common concern of humankind* has strong conceptual and legal significance, but a corpus hit for that phrase cannot simply be counted as an instance of fear. Likewise *vulnerability*, *risk*, *threat*, and *loss and damage* identify possible or experienced harms; none independently demonstrates a felt affect. ³⁹

The especially important lexical development is therefore not “more fear” but the **formalisation of temperature itself as a thresholded governance object**: *global average temperature*, *2°C*, *1.5°C*, and *pre-industrial levels* become linked within the Agreement’s goal structure. ⁵

Anchor — 2022

Historical context. WMO’s provisional 2022 state-of-climate report described the preceding eight years as on course to be the eight warmest on record and explicitly referred to *ever-rising greenhouse gas concentrations*, *accumulated heat*, and *extreme heatwaves*. The later final 2022 report likewise emphasised heat-trapping greenhouse gases and global temperature. ⁴³

By this period, psychological and affective vocabulary is much easier to establish directly. Hickman et al.’s global youth survey is explicitly titled with *climate anxiety* and reports worry and other climate-related emotions; a 2022 Frontiers scoping review explicitly discusses *eco-anxiety*, *psychological distress*, worry, and hopelessness in anticipation of climate change. These modern constructs are analytically useful precisely because they can now be documented directly rather than inferred retrospectively. ⁴⁴

At the same time, public and institutional discourse around 2022 contains high-intensity threat frames including *climate crisis* and *climate emergency*. UN-linked 2022 materials explicitly use climate-emergency language, while contemporary UN-related communication describes the threat posed by the climate crisis. ⁴⁵

Evidence key. **E22-1** = WMO, *Provisional State of the Global Climate in 2022*. ⁴⁶ **E22-2** = WMO, *State of the Global Climate 2022*. ⁴⁷ **E22-3** = IPCC AR6 WGII materials concerning climate-related heat risk and impacts. ⁴⁸ **E22-4** = Hickman et al., global youth climate-anxiety study, published at the end of 2021 and therefore used as near-anchor evidence for vocabulary available by 2022. ⁴⁹ **E22-5** = 2022 Frontiers scoping review on eco-anxiety in children. ⁵⁰ **E22-6** = UN/Geneva Environment Network 2022 climate-crisis/emergency material. ⁴⁵ **E22-7** = contemporary reporting of the UN Secretary-General's "climate catastrophe" formulation; retained with lower confidence than official-source terms. ⁵¹ **E22-N** = current Google Books Ngram Viewer interface, used solely to verify the 2022 endpoint. ⁹

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
2022-A-01	2022	global temperature	A	phrase	planetary temperature	2022	E22-1,2	WMO global temperature indicator
2022-A-02	2022	global mean temperature	A	phrase	global average temperature	2022	E22-2	WMO global temperature
2022-A-03	2022	rising temperatures	A	phrase	warming	2022 context	E22-1–3	Rising temperature frame established across climate materials
2022-A-04	2022	warming	A	unigram	temperature increase	2022	E22-2,3	Standard climate process
2022-A-05	2022	global warming	A	phrase	planetary warming	2022	E22-2,3	Established climate
2022-A-06	2022	extreme heat	A	phrase	severe heat	2022	E22-1–3	Contemporary climate reporting promotes treatment of heat
2022-A-07	2022	extreme heatwaves	A	phrase	severe heat-wave events	2022	E22-1	Exact event

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
2022-A-08	2022	ocean heat	A	phrase	ocean thermal content	2022	E22-2	WMO indicates inclusion of heat
2022-B-01	2022	climate change	B	phrase	—	2022	E22-1–3	Dominant scientific institution term
2022-B-02	2022	human-induced climate change	B	phrase	anthropogenic climate change	2022	E22-3	IPCC framework
2022-B-03	2022	climate system	B	phrase	Earth climate system	2022	E22-3	Standard framework
2022-B-04	2022	greenhouse gases	B	phrase	GHGs	2022	E22-1,2	Atmosphere driven reports
2022-B-05	2022	heat-trapping greenhouse gases	B	phrase	radiatively active GHGs	2022	E22-2	WMO uses trapping form
2022-B-06	2022	climate-related	B	morphological form	connected to climate	2022	E22-2,3	Wide modifier climate assessments
2022-B-07	2022	global climate	B	phrase	planetary climate	2022	E22-1,2	WMO scale
2022-B-08	2022	changing climate	B	phrase	climate change	2022	E22-1–3	Contextual variation remains available
2022-C-01	2022	climate anxiety	C	phrase	anxiety about climate change	by 2021–22	E22-4	Explicit scholarly construction vocabulary
2022-C-02	2022	eco-anxiety	C	phrase	environmentally oriented anxiety/distress	2022	E22-5	Explicit 2022 literature

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
2022-C-03	2022	worried about climate change	C	phrase	climate worry	by 2021–22	E22-4	Survey opera worry clima
2022-C-04	2022	worry	C	unigram	worry	by 2021–22	E22-4,5	Explic ident nega emot react
2022-C-05	2022	psychological distress	C	phrase	distress	2022	E22-5	Review descr signifi psych distre clima aware
2022-C-06	2022	afraid	C	morphological variant	fear	by 2021–22	E22-4	Clima resea recor afraid clima emot
2022-C-07	2022	hopelessness	C	unigram	hopeless affect	2022	E22-5	Eco-a litera explic includ hope
2022-D-01	2022	risk	D	unigram	risk	2022	E22-2,3	AR6/ organ impa throu
2022-D-02	2022	climate-related risks	D	phrase	climate risks	2022	E22-2,3	Offici asses vocabl explic assoc with
2022-D-03	2022	threat	D	unigram	threat	2022	E22-6	Instit public compr ident clima as th

Candidate ID	Anchor	Candidate term	Layer	Type	Normalised concept	Historical attestation	Evidence source	Historical evidence
2022-D-04	2022	climate crisis	D	phrase	climate framed as crisis	2022	E22-6	Explicit context expression
2022-D-05	2022	climate emergency	D	phrase	climate framed as emergency	2022	E22-6	Explicit context UN-like vocabulary
2022-D-06	2022	climate catastrophe	D	phrase	catastrophic climate outcome	2022	E22-7	Context UN-w/ disaster catastrophe form
2022-D-07	2022	devastating	D	unigram	severe destructive impact	2022	E22-1	WMO description destructive context extreme

Major ambiguity issues. Modern affect vocabulary is better attested, but it remains differentiated. *Climate anxiety*, *eco-anxiety*, *worry*, *afraid*, *psychological distress*, and *hopelessness* cannot be collapsed into one emotion variable without semantic work. Hickman et al.'s study itself deals with a range of emotional reactions rather than one undifferentiated state, and the 2022 eco-anxiety review similarly groups multiple responses around an emerging construct. ⁵²

Likewise, *extreme heat* and *extreme heatwaves* are physical/event expressions, *climate emergency* and *climate crisis* are threat frames, while *climate anxiety* names an affective/psychological phenomenon. Their co-presence in 2022 does not establish that they are interchangeable.

Corpus endpoint clarification. 2022 is the endpoint used for corpus-oriented design because the currently available Google Books Ngram Viewer permits searches through 2022. August 2026 is the present research date, not the Google Books endpoint. No Ngram frequency result was consulted in this Round 1 study. ⁹

Cross-Anchor Synthesis, Round Two Package, and Full References

Descriptive candidate inventory

The table below reports only relationships **inside this 180-record exploratory inventory**. “First validated here” means first among these six sampled positions; it does not claim historical coinage or first-ever use.

Descriptive pattern	Examples	Methodological meaning
Early observational vocabulary	<i>temperature; mean temperature; high temperature; heat; climate; meteorological observations; weather</i>	Already available around 1842; should not be redescribed using later climate-crisis terminology
Older lexical forms retained or transformed	<i>change of climate; climatic change; artificial production of carbon dioxide; human agency</i>	Historically specific surface forms should be queried as written rather than silently modernised
CO ₂ -temperature causal configuration	<i>carbon dioxide; atmospheric carbon dioxide; temperature increase; climatic change; human agency</i>	Strongly evidenced by the Callendar bridge without corresponding strong fear vocabulary
Institutional climate-risk configuration	<i>climate change; global warming; greenhouse gases; concern; threatening; disastrous; global security</i>	Strong at 1988; affect and threat must nevertheless remain distinct
Public high-intensity threat configuration	<i>risk; dangerous; worst impacts; climate crisis; planetary emergency</i>	Strongly documented in 2006–07 materials
Temperature-threshold configuration	<i>global average temperature; 2°C; 1.5°C; pre-industrial levels</i>	Particularly diagnostic of 2015 governance language
Explicit modern climate-affect configuration	<i>climate anxiety; eco-anxiety; worried about climate change; psychological distress; afraid; hopelessness</i>	Directly documentable around 2022; not licensed for backward projection
Recurring but semantically unstable terms	<i>temperature; heat; climate; concern; risk; threat; warming</i>	Recurrence is descriptive only and does not establish semantic continuity

A few cross-anchor recurrences deserve preservation for later semantic analysis. *Climate* appears from the early baseline onward, but nineteenth-century climatology and contemporary climate-change governance are not the same conceptual environment. *Climatic change* is useful in 1949 and 1988, while *climate change* is institutionally central by 1988 and thereafter. *Global warming* is strongly validated at 1988, 2006–07, 2015, and 2022 in this inventory. *Risk* becomes strongly usable from the later institutional/public-policy record, particularly 2006 onward. *Concern* is present institutionally before modern climate-psychology terminology but remains semantically ambiguous. ⁵³

Exclusion and ambiguity register

Term investigated	Decision in this round	Reason
climate anxiety at 1842	Exclude as anachronistic	No evidence supports projecting the modern compound into the nineteenth-century baseline
eco-anxiety at 1842/1938/1988	Exclude	Modern environmental-psychological construct; direct period attestation not established

Term investigated	Decision in this round	Reason
global warming at 1842	Exclude	Modern global anthropogenic-warming phrase would distort the period's observational lexicon
climate crisis at 1842/1938	Exclude	Not validated in the historical source sets
climate emergency at early anchors	Exclude	Contemporary high-intensity threat framing cannot be silently back-projected
depression	Exclude from core inventory	Psychiatric, economic, meteorological and topographical senses make it especially hazardous without contextual filtering
alarm	Conditional/No at several anchors	Can denote an emotion, an alert, a warning system, or generic alarm; direct climate-affective use must be shown
anxiety	No at early/middle anchors; Yes in <i>climate anxiety</i> at 2022	Bare word is broad; modern compound has direct scholarly attestation
concern	Conditional in many settings	Can be affective, epistemic, diplomatic, legal, or simply "a matter of concern"
threat	Layer D only	Threat appraisal is not itself evidence of fear
risk	Layer D only	Probabilistic/institutional framing is not synonymous with affect
crisis	Layer D only	Generic political/economic crises create false positives; climate modifier is important
heat	Retain with strong ambiguity flag	Can refer to weather, body temperature, cooking, machinery, combustion, competition, policing metaphor, and more
warming	Retain from later anchors with ambiguity flag	Can describe any object/process warming; climate context needed
climate	Retain throughout with ambiguity flag	Historical scientific meanings differ, and metaphorical/social uses are common
change	Exclude as standalone keyword	Excessively unconstrained; use historically attested phrases such as <i>climatic change</i> or <i>climate change</i>
public awareness	Attested but not Layer C	Cognitive/communication concept rather than affect under this thesis's narrow definition
urgency	Usually Conditional or excluded from C	Primarily temporal/action orientation; does not independently establish emotion

Term investigated	Decision in this round	Reason
vulnerability	Conditional Layer D	Susceptibility to harm rather than threat or fear itself
adaptation/ resilience/mitigation	Not expanded as lexical layers	Outside Round 1 A–D boundary except where necessary to contextualise official sources
heritage/memory	Excluded from candidate layers	Belongs to the broader thesis framing, not this lexical-discovery round

Bridge-anchor recommendations and resulting timeline

1938–1949: ACCEPT — specific year: 1938. The strongest reason is lexical rather than commemorative. Callendar's 1938 title and analysis bring human-produced CO₂ and temperature into one causal proposition. 1949 strengthens the bridge through *climate*, *climatic change*, *temperature rise*, and *human agency*, but the earlier date is the cleaner anchor. The bridge improves the design because otherwise the timeline would jump directly from nineteenth-century observational climate language to 1988's already mature institutional global-warming vocabulary. ⁵⁴

2006–2007: ACCEPT — period. 2006 is especially valuable for Stern's *risk/danger/impacts* configuration and Gore's *climate crisis* communication; 2007 adds AR4's scientific consolidation and the Nobel knowledge-dissemination moment. Neither alone represents the full documentary transition under investigation. ³⁷

Recommended six-position design:

1842 → 1938 → 1988 → 2006–2007 → 2015 → 2022

The associated labels should be:

1842: contextual environmental-history / meteorological baseline

1938: CO₂–temperature causal-science bridge

1988: institutional climate-change and risk anchor

2006–2007: public-communication / threat-framing bridge

2015: temperature-threshold global-governance anchor

2022: contemporary heat/risk/affect corpus endpoint

Input package for Deep Research Round Two

The following contains every record marked **Yes** or **Conditional** above. “Carry Forward” preserves uncertainty; it is not a claim that the candidate will survive semantic validation.

Anchor	Term	Layer	Confidence	Carry Forward
1842	temperature	A	Strong	Yes
1842	mean temperature	A	Strong	Yes
1842	high temperature	A	Strong	Yes
1842	heat	A	Strong	Yes

Anchor	Term	Layer	Confidence	Carry Forward
1842	oppressive heat	A	Weak	Conditional
1842	too hot and close	A	Strong	Yes
1842	climate	B	Strong	Yes
1842	meteorology	B	Strong	Yes
1842	meteorological observations	B	Strong	Yes
1842	weather	B	Strong	Yes
1842	atmosphere	B	Strong	Yes
1842	seasons	B	Strong	Yes
1842	change of climate	B	Moderate	Conditional
1842	fear	C	Moderate	Conditional
1842	relief	C	Weak	Conditional
1842	serious importance	D	Strong	Conditional
1842	illness	D	Moderate	Conditional
1842	death	D	Strong	Yes
1842	loss	D	Strong	Conditional
1842	damage	D	Strong	Conditional
1842	heat ... caused the death	D	Strong	Yes
1938	temperature	A	Strong	Yes
1938	mean temperature	A	Strong	Yes
1938	world temperatures	A	Strong	Yes
1938	temperature increase	A	Strong	Yes
1938	rise in temperature	A	Strong	Yes
1938	surface temperatures	A	Strong	Yes
1938	temperature observations	A	Strong	Yes
1938	temperature change	A	Strong	Yes
1938	carbon dioxide	B	Strong	Yes
1938	atmospheric carbon dioxide	B	Strong	Yes
1938	artificial production of carbon dioxide	B	Strong	Yes
1938	atmosphere	B	Strong	Yes
1938	climate	B	Strong	Yes
1938	climatic change	B	Strong	Yes

Anchor	Term	Layer	Confidence	Carry Forward
1938	carbon dioxide theory	B	Moderate	Conditional
1938	human agency	B	Strong	Yes
1938	doubtful	C	Moderate	Conditional
1938	uncertainty	C	Moderate	Conditional
1938	aridity	D	Strong	Conditional
1938	recession of the glaciers	D	Strong	Conditional
1988	global warming	A	Strong	Yes
1988	warming	A	Strong	Yes
1988	climate warming	A	Strong	Yes
1988	global temperature	A	Moderate	Yes
1988	temperature rise	A	Moderate	Conditional
1988	temperature increase	A	Moderate	Yes
1988	warming trend	A	Moderate	Conditional
1988	summer heat waves	A	Moderate	Yes
1988	climate change	B	Strong	Yes
1988	climatic change	B	Strong	Yes
1988	global climate	B	Strong	Yes
1988	global climate patterns	B	Strong	Yes
1988	greenhouse gases	B	Strong	Yes
1988	atmospheric concentrations	B	Strong	Yes
1988	human activities	B	Strong	Yes
1988	changing atmosphere	B	Strong	Yes
1988	concern	C	Strong	Yes
1988	concerned	C	Strong	Yes
1988	urgency	C	Moderate	Conditional
1988	threatening	D	Strong	Yes
1988	severe consequences	D	Strong	Yes
1988	disastrous	D	Strong	Yes
1988	detrimental effects	D	Strong	Yes
1988	critical problem	D	Strong	Yes
1988	global security	D	Strong	Yes

Anchor	Term	Layer	Confidence	Carry Forward
1988	ultimate consequences	D	Strong	Yes
2006-07	global warming	A	Strong	Yes
2006-07	warming	A	Strong	Yes
2006-07	global average temperature	A	Strong	Yes
2006-07	temperature increase	A	Strong	Yes
2006-07	temperature rise	A	Strong	Yes
2006-07	warming of the climate system	A	Strong	Yes
2006-07	heat waves	A	Strong	Yes
2006-07	warmer	A	Moderate	Conditional
2006-07	climate change	B	Strong	Yes
2006-07	climate system	B	Strong	Yes
2006-07	greenhouse gases	B	Strong	Yes
2006-07	greenhouse gas concentrations	B	Strong	Yes
2006-07	anthropogenic warming	B	Strong	Yes
2006-07	human influence	B	Strong	Yes
2006-07	radiative forcing	B	Strong	Yes
2006-07	global climate	B	Moderate	Yes
2006-07	concern	C	Moderate	Conditional
2006-07	urgency	C	Moderate	Conditional
2006-07	risk	D	Strong	Yes
2006-07	dangerous	D	Strong	Yes
2006-07	serious impacts	D	Strong	Yes
2006-07	worst impacts	D	Strong	Yes
2006-07	climate crisis	D	Strong	Yes
2006-07	planetary emergency	D	Moderate	Yes
2006-07	threat	D	Strong	Yes
2015	global average temperature	A	Strong	Yes
2015	temperature increase	A	Strong	Yes
2015	global temperature rise	A	Strong	Yes
2015	2°C	A	Strong	Yes
2015	1.5°C	A	Strong	Yes

Anchor	Term	Layer	Confidence	Carry Forward
2015	global warming	A	Strong	Yes
2015	extreme heat	A	Strong	Yes
2015	heat waves	A	Strong	Yes
2015	climate change	B	Strong	Yes
2015	greenhouse gas emissions	B	Strong	Yes
2015	pre-industrial levels	B	Strong	Yes
2015	climate system	B	Moderate	Yes
2015	anthropogenic emissions	B	Strong	Yes
2015	global response	B	Strong	Yes
2015	low GHG emissions	B	Strong	Yes
2015	changing climate	B	Strong	Yes
2015	concern	C	Moderate	Conditional
2015	common concern of humankind	C	Moderate	Conditional
2015	threat of climate change	D	Strong	Yes
2015	risk	D	Strong	Yes
2015	adverse effects	D	Strong	Yes
2015	loss and damage	D	Strong	Yes
2015	vulnerability	D	Strong	Conditional
2015	danger	D	Strong	Yes
2015	present threat	D	Strong	Yes
2022	global temperature	A	Strong	Yes
2022	global mean temperature	A	Strong	Yes
2022	rising temperatures	A	Moderate	Yes
2022	warming	A	Strong	Yes
2022	global warming	A	Strong	Yes
2022	extreme heat	A	Strong	Yes
2022	extreme heatwaves	A	Strong	Yes
2022	ocean heat	A	Strong	Yes
2022	climate change	B	Strong	Yes
2022	human-induced climate change	B	Strong	Yes
2022	climate system	B	Strong	Yes

Anchor	Term	Layer	Confidence	Carry Forward
2022	greenhouse gases	B	Strong	Yes
2022	heat-trapping greenhouse gases	B	Strong	Yes
2022	climate-related	B	Moderate	Conditional
2022	global climate	B	Strong	Yes
2022	changing climate	B	Strong	Yes
2022	climate anxiety	C	Strong	Yes
2022	eco-anxiety	C	Strong	Yes
2022	worried about climate change	C	Strong	Yes
2022	worry	C	Strong	Yes
2022	psychological distress	C	Strong	Yes
2022	afraid	C	Strong	Yes
2022	hopelessness	C	Moderate	Yes
2022	risk	D	Strong	Yes
2022	climate-related risks	D	Strong	Yes
2022	threat	D	Strong	Yes
2022	climate crisis	D	Strong	Yes
2022	climate emergency	D	Strong	Yes
2022	climate catastrophe	D	Moderate	Conditional
2022	devastating	D	Strong	Conditional

The Round Two package therefore preserves **147 of the 180 exploratory records** as either Yes or Conditional; the remaining 33 are retained in the full Round One tables as negative or failed evidence. This is an inventory decision, not a quantitative statement about language prevalence.

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Round One quality-control result. Exactly **six chronological positions** have been examined and exactly **30 records per position** are reported: **180 anchor-specific candidate records in total**, with each anchor constructed as **8 Layer A + 8 Layer B + 7 Layer C + 7 Layer D**. The design preserves weak

and failed candidates rather than hiding them; treats 1842 as an evaluated baseline rather than an origin myth; recommends 1938 as the specific first bridge and 2006–2007 as the second bridge period; separates 2022 from the 2026 research present; and makes no claim that recurrence proves semantic continuity, that threat proves fear, that any historical population was “more afraid,” or that fear itself has already been demonstrated to constitute cultural heritage.

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